

ENOCK KUMI ACKAAH

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EDUCATION

M.S. in Applied Statistics, Biostatistics <i>Department of Mathematics & Statistics / Georgia State University</i>	Expected May 2028 <i>Atlanta, GA</i>
M.S. in Statistics <i>Department of Statistics / The University of Akron</i>	May 2026 <i>Akron, OH</i>
B.Sc. in Mathematics and Statistics <i>School of Physical Sciences / University of Cape Coast</i>	August 2023 <i>Cape Coast, Ghana</i>

RESEARCH INTERESTS

- **Biomedical Machine Learning:** cancer prediction, high-dimensional omics, RNA sequencing, DNA methylation, and predictive modeling for biomedical data.
- **Explainable and Trustworthy AI:** model reliability, external validation, calibration, reproducibility, feature stability, and clinical generalizability.
- **Applied Statistical Learning:** regression modeling, classification, survival analysis, multivariate methods, statistical consulting, and data-driven decision support.

PUBLICATIONS & MANUSCRIPTS

- Akosile, C., Ezenwanne, O., **Ackaah, E.**, Ajieh, F., et al. “Predictive Analytics and Machine Learning for Energy Forecasting and Load Optimisation in Hospital Systems: A Critical Narrative Review.” *Journal of Scientific Research and Reports*. doi:10.9734/jsrr/2026/v32i94483, 2026.
- Ibrahim, M. S., **Ackaah, E.**, Acheampong, C. B., Arhin, E., et al. “Reducing False Positives and False Negatives in Artificial Intelligence-assisted Imaging Screening: A Critical Review of Accuracy, Error Mechanisms and Clinical Translation.” *Journal of Advances in Medical and Pharmaceutical Sciences*. doi:10.9734/jamps/2026/v28i9891, 2026.
- **Ackaah, E. K.**, & Datta, S. “Cancer Classification in Low- and High-Dimensional Biomedical Data Using Machine Learning Models.” *Preprints*. <https://doi.org/10.20944/preprints202608.0073.v1>, 2026.
- **Ackaah, E. K.**, & Datta, S. “Artificial Intelligence in the Biomedical Sciences and Healthcare—Its Impact, Potential Benefits and Pitfalls.” Manuscript in preparation for submission to *Precision Oncology*, 2026.
- **Ackaah, E. K.** “Statistical Reliability of High-Dimensional ER-Status Prediction Models” Version 0.1.1. [Computer software]. Zenodo. doi:10.5281/zenodo.21524693, 2026.
- **Ackaah, E. K.** “Frozen 72-CpG DNA Methylation Classifier for Colorectal Tumor Detection” Version 1.0.0. [Computer software]. Zenodo. doi:10.5281/zenodo.21819138, 2026.

RESEARCH AND OTHER PROJECTS

Lung Cancer Prediction Using Machine Learning Models <i>Graduate Machine Learning Coursework Project / The University of Akron</i>	December 2025 <i>Akron, OH</i>
• Applied multiple machine learning models to a publicly available Kaggle lung cancer dataset. • Implemented data preprocessing, model fitting, and performance evaluation using logistic regression, decision trees, random forests, support vector machines, and neural networks. • Evaluated model performance using AUC, RMSE, ROC curves, and confusion matrices; presented findings in a graduate statistics course presentation.	
Dimensionality Reduction Using PCA and Multidimensional Scaling <i>Graduate Multivariate Statistics Project / The University of Akron</i>	December 2025 <i>Akron, OH</i>
• Applied Principal Component Analysis to the Iris dataset and Multidimensional Scaling to Framingham Heart Study cardiovascular risk-factor data. • Used PCA to reduce four correlated Iris measurements to two principal components explaining approximately 95.81% of total variance. • Used MDS to visualize similarity structure among cardiovascular risk profiles and identify patterns related to ten-year coronary heart disease events.	

Smartphone Battery Life Experimental Design and ANOVA Project

December 2025

The University of Akron

Akron, OH

- Conducted under the supervision of Dr. Richard Einsporn.
- Designed and analyzed a Generalized Randomized Complete Block Design to study the effects of charging time, app usage, and smartphone brand on battery drop percentage.
- Used ANOVA, residual diagnostics, interaction plots, and Tukey pairwise comparisons to evaluate factor effects and compare treatment means.

WORK EXPERIENCE

Graduate Teaching Assistant

August 2026 – Present

Georgia State University

Atlanta, GA

- Assisted students in Elementary Statistics using Microsoft Excel.
- Conducted online tutoring sessions for students in the Commons MILE Lab.
- Assisted undergraduate students with College Algebra in the MILE Lab.

Graduate Teaching Assistant

August 2024 – May 2026

The University of Akron

Akron, OH

- Led weekly statistics laboratories for approximately 15–20 students, integrating statistical concepts with software-based analysis.
- Tutored students in statistics, probability, R, SPSS, Minitab, and quantitative problem solving.
- Graded assignments and assessments for large sections and provided detailed feedback to support learning.

TECHNICAL SKILLS

Programming: Python, R, C, Sklearn, Numpy, Pandas, PyTorch, Tensorflow, Huggingface

Tools: Git, Linux, Jupyter, VS Code

Skills: Large Language Models, Trustworthy AI, Data Analysis, Statistical Learning Theory

PROFESSIONAL MEMBERSHIPS

- Student Member, **American Statistical Association** 2026–Present